Introduction:

Patella femoral pain is a widespread problem affecting ~23% of the population (1), a leading cause of patella femoral pain is patella maltracking. Previously, the design specification and preliminary design for a medical device was established, using stereo cameras to capture and track the 3D position of the patella and provide quantitative tracking data (2). In order to be lawful, and by extension marketable, medical devices must demonstrate that they are safe and perform as intended. In the United Kingdom, medical devices must comply with Medical Devices Regulations 2002 (3) prior to being placed on the market. These regulations define device classification, safety, performance and management requirements. In addition, device compliance is supported by recognised international standards, particularly ISO 14971, detailing medical device risk management throughout the lifecycle of the device.

A structured risk management approach is essential. Failure Modes and Effects Analysis (FMEA) is a systematic method for identifying potential failure modes, their causes, and the resulting hazards and harms. When applied in accordance with ISO 14971, it provides a framework for risk identification, evaluation and ultimately reduction.

This paper details application of relevant medical device regulations and an FMEA-based risk management process to the design specification of a knee-mounted patellar tracking device, to develop a functional and legally compliant device for clinical use.

Regulatory scope and device classification:

The United Kingdoms' Medical Device Regulation 2002 provides a broad range of requirements for medical devices, the requirements of particular interest for this project are found in section II. This regulation is derived from European medical device legislation, specifically Directive 93/42/EEC (4), which outlines requirements within its Annexes.

Annex IX provides methods for device classification. Using this framework the device can be classified as a Class I non-invasive diagnostic medical device. It is considered transient as its use does not exceed 60 minutes. Annex II places requirements on construction and environmental properties, including minimising risks arising from physical features, providing ergonomic fit, and ensuring safe operation under foreseeable environmental conditions such as electromagnetic interference, electrostatic discharge, and temperature variation. Ageing of materials and potential loss of measurement accuracy over time must also be considered, as these could compromise safety or performance.

Because the device performs a measuring function, Annex I further requires that it provides sufficient accuracy for its intended purpose and clearly states the limits of that accuracy. This directly informs the need for calibration procedures, accuracy verification, and clear specification of measurement uncertainty within the device documentation.

Finally, as the device is connected to an internal energy source, additional requirements apply to minimise electromagnetic emissions and reduce the risk of electric shock or thermal injury. All the considerations are evaluated systematically in the FMEA table. Other annexes of Directive 93/42/EEC, including those relating to notified body certification, production quality assurance, and full clinical evaluation, are not performed within the scope of this project. However, awareness of these annexes informs future steps required should the device progress beyond a prototype toward commercial deployment.

ISO 14971: 2019 + A11:2021 risk management process

ISO 14971 (5) sets out a risk management framework for medical devices, requiring manufacturers to identify hazards, estimate and evaluate risk, implement risk controls, and monitor the ongoing effectiveness of those controls throughout the device's lifecycle. The standard expects a documented, systematic, auditable process, making FMEA an appropriate tool for structuring risk analysis.

Figure A1 provides a diagrammatic representation of the risk management process. The process begins with a risk management plan, defining the intended use, reasonably foreseeable misuse, responsibilities, risk acceptability criteria, and the methods used for risk estimation and evaluation. A risk analysis is then performed to identify hazards and the foreseeable sequences of events that could lead to the hazards developing into harm. Risk estimation must account for both the severity of harm and the probability of occurrence. In Table II, severity (S) and probability (P) are rated from 1 to 10 and multiplied to calculate the grade (G).

Following analysis, risk evaluation determines whether each estimated risk is acceptable against the predefined criteria, in this project a graded risk < 18 is deemed acceptable. Where risks are not acceptable, risk control measures must be implemented using the prescribed hierarchy: inherently safe design and manufacture; protective measures in the device or manufacturing process; information for safety and training. Finally, an overall residual risk evaluation must be performed, judging whether the remaining residual risks are acceptable when weighed against the device's intended clinical benefits; if unacceptable, any significant residual risks must be communicated to users through labelling and instructions for use.

Design Revisions & Justifications:

Following the evaluation, as seen in table III, key design changes were identified and implemented. A silicone rubber gasket was integrated between the enclosure halves to provide IP45 ingress protection, limiting exposure to dust and protecting against water splashes from all direction, this due to the high likelihood of liquid ingress during cleaning or as a result of perspiration, which could lead to internal corrosion or short circuits. Silicone was selected due to its chemical resistance, elasticity, and long-term sealing performance under repeated compression (6,7).

The brace material was change from Fused Deposition Modelled (FDM) Poly(Lactic Acid) (PLA) to Selective Laser Sintered (SLS) PA12 Nylon due to its superior aging properties. Table I compares key material properties of interest for the chosen application. PA12 Nylon has an equivalent ultimate tensile strength (UTS) to PLA while having a significantly reduced Young's modulus, increasing its strain capacity by a factor of ~2, and improving its resistance to brittle fracture under cyclic loading. Additionally, both materials have high electrical resistance and will act as electrical insulators, decreasing the risk of current being transferred to the user. The main advantage of PA12 Nylon is its chemical stability, displaying limited degradation due to aging, cleaning or moisture exposure, making it more suited to PLA for long term use in a clinical environment.

Table I: Material Comparison

Material	Youngs Modulus	UTS	Aging	Cleaning	Electrical resistance	Cost
Nylon PA12	1650 MPa (8)	48 MPa (8)	Good environmental stability (9)	Limited degradation with common cleaners showing low absorption and degradation (10)	$6.7 \times 10^{14}\Omega$ (11)	Higher cost (8)
FDM PLA	3039 MPa (12)	48 MPa (12)	Prone to embrittlement, Damage under UV exposure. hydrolysis during humid conditions (13,14)	Degradation from common cleaning products. (15)	$1.16 \times 10^{14}\Omega$ (11)	Ultra Low cost ~ £10 (16)

The previous design specification did not utilise mathematical models to determine whether the material could withstand the forces applied. *Figure A2* represents a free body diagram of a supporting strut of the brace, with the hinge modelled as a fixed support. The strut is acted on by a lateral force, at distance x, where x is the length of the strut. Using equation 4, the force required to fracture the strut can be calculated.

$$F = \frac{48 \times 10^{6} \cdot 12.7 \times 10^{-3} \cdot (25.4 \times 10^{-3})^{2}}{6 \times 150 \times 10^{-3}} \approx 437N$$

Using safety factor of 2:

$$F \approx 218.5N$$

The applied force only comes from a valgus/ varus moment at the brace; these forces *in vivo* are a fraction of bodyweight forces and estimated to be below 100N. This is less than the calculated force of 218.5N, verifying the safety of nylon PA12 for use. (3)

The internal battery was removed as its accumulated electrical risks exceeded acceptable thresholds. Primary risk mitigation under ISO 14971, is inherently safe design, which can be achieved by total removal of the hazard. As data collection is conducted using a wired connection, this change does not significantly increase trip or snag hazards. Furthermore, chamfered and filleted edges were applied throughout the brace to reduce injury risk, and disposal and alignment instructions were added. Collectively, these changes improve device safety, robustness, and accuracy during clinical use.

Conclusion:

After reviewing medical regulation, the device was classified as a class I medical device per directive 93/42/EEC as referenced by the United Kingdom Medical Device Regulation 2002. Following ISO 14971 risk management for medical devices, an FMEA table was constructed to quantitatively analyse hazards, graded using their severity and probability of occurrence while also envisaging the reasonably forceable scenarios in which these hazards could develop. Risks graded above 18 were deemed to be above the acceptable risk tolerance. 13 risks exceeded this requirement; these were then actioned upon by revising the design. The most impactful changes were the removal of the internal battery, in favour of external power supply, changing the brace and housing material from poly-lactic acid to PA12 Nylon and the inclusion of a silicon gasket to provide IP 45 water and dust resistance, preventing liquid ingress into the electronics housing.

Table II: ISO 14971 FMEA

Item	Lifecycle Stage	Hazard	Hazard location	Initiating circumstance	Risk/Harm	S	P	G
Brace hinge assembly	Use	Brace hinge breaks	Hinge joint at transepicondylar axis	High load, impact, fatigue, print defect, incorrect fastener torque	Sudden loss of support, fall risk, sharp edges, invalid measurement	9	2	**18**
Brace frame	Use	Brace cracks/fracture	Struts near hinge, strap anchors	Low infill, layer delamination, stress concentration, ageing	Sharp edges, loss of camera pose	8	2	**16**
Components	Use	High bending moment	Camera mount, ` electronics housing	High weight off axis	Change in user gait	9	2	**18**
Fasteners	Assembly / Use	Screw loosens and parts detach	Camera mount, electronics housing	Vibration, no locking feature	Inaccurate results, electric shocks	6	2	**12**
Velcro straps	Use	Strap loosens causing migration	Thigh/calf straps	Sweat, wear, poor tensioning	Loss of alignment, data error	2	6	**12**
Velcro straps	Use	Over-tightening pressure injury	Strap contact regions	Clinician error	Pain, bruising, impaired circulation	3	4	**12**
Brace interface	Use	Skin rubbing/irritation	Strap zones, brace edges	Repeated motion, sweat	Dermatitis, blistering	4	5	**20**
Brace interface	Use	Allergic reaction	Skin-contact surfaces	Material sensitivity	Rash, swelling	3	1	**3**
Brace interface	Cleaning / Use	Cross-infection	All contact surfaces	Inadequate cleaning	Infection transmission	6	5	**30**
Brace material	Storage / Use	Material ageing/embrittlement	High-stress regions, e.g hinge	UV, heat, cleaning chemicals	Increased fracture risk	5	7	**35**
Brace material	Use	Flammability	Near electronics	Electrical overheating	Burns, fire spread	10	1.5	**15**
Hinge alignment	Setup / Use	Misalignment to anatomical axis	Hinge vs TEA	Landmark palpation error	Systematic measurement error	4	5	**20**
Potentiometer	Use	Electrical drift/noise	Internal tracks	Wear, contamination	Angle jitter	4	4	**16**
Camera mount arm	Use	Mount fracture/bending	Camera support arm	Impact, fatigue	Camera detachment, or change in orientation	7	2	**14**
Camera	Use	Patella exits field of view	Optical path	Migration, posture	Data gaps	7	2	**14**
Camera	Use	Minimum distance violated	Depth sensing	Excess flexion, anatomy	Depth dropout	8	2	**16**
Camera	Use	Optical occlusion	Patella region	Soft tissue, clothing	Tracking loss	8	1	**8**
Camera	Use	Environmental interference	Stereo cameras	Lighting glare, insufficient lighting, sweat	Depth failure	8	4	**32**
Electronics housing	Use	Sharp enclosure edges	Housing perimeter	Harsh angles in design	Cuts, discomfort	5	6	**30**
Electronics	Use	Overheating	ESP32, battery	High current	Burns, shutdown	8	2	**16**
Battery system	Use	Electrical fire	Battery pack	Internal short	Burns, smoke	10	1	**10**
Battery system	Use	Battery leakage/venting	Battery pack	Damage, overheating	Chemical burns	8	1	**8**
Charging	Maintenance	Unsafe charging	Charging port	Incorrect charging, electrical fault	Fire, electric shocks	9	1	**9**
Wiring	Use	Cable snag/trip hazard	External leads	Gait movement	Falls	3	6	**18**
Wiring	Use	Electrical shock	Exposed wiring	Insulation damage	Shock	9	2	**18**

Liquid ingress	Use / Cleaning	Short circuit	Ports, PCB	Sweat, fluids	Shock, fire	8	3	**24**
Software	Use	Tracking failure	Processing pipeline	Occlusion, anatomy	Wrong output	7	4	**28**
Software	Use	Time synchronisation error	Data streams	Dropped frames	Incorrect plots	5	4	**20**
UI / Output	Use	Graph misinterpretation	Display	Poor labelling	Wrong clinical action	6	2	**12**
Labelling / IFU	Use	Use outside intended population	Clinical context, patient	High BMI patients, prolonged use, broken skin	Invalid results	4	4	**16**
Disposal	End-of-life	Unsafe battery disposal	Battery pack	No guidance	Fire risk	5	5	**25**
Manufacturing	Manufacture	Dimensional inaccuracy	Hinge/mount	Warping	Misfit, loosening	7	3	**21**

Table III: Actions and Evaluation

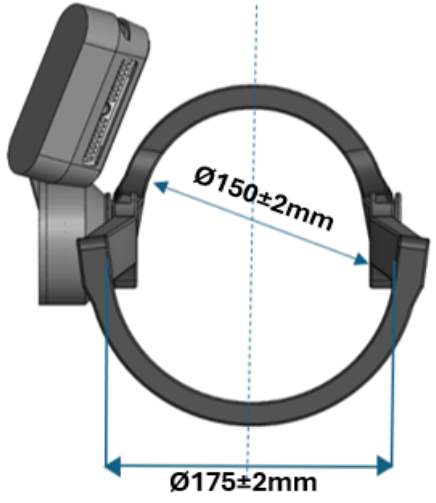
Item	Actions recommended	Actions taken	R.S	R.P	R.G
Brace interface	Post processing to ensure a smooth finish	Media blasting and polishing process	3	3	**9**
Components	Decrease mass of components	Battery removed, replaced with wired connection, camera position changed to front view instead of side view.	4	1	**4**
Brace interface (cleaning)	Cleaning protocol, wipe-compatible materials	Material changed to PA12 nylon	6	1	**6**
Brace material	Change material to PA12 nylon	As stated	3	2	**6**
Hinge alignment	Alignment marks, setup checklist	Provide instruction on alignment techniques	2	2.5	**5**
Camera	Detailed instruction	As stated	7	2	**12**
Electronics housing	Chamfers, finishing process	As stated	2	2	**4**
Liquid ingress	IPX5 requirements	Rubber gasket	8	1	**8**
Software	Confidence metrics	As stated	4	4	**16**
Software	Time stamp verification method	Provide a method for synchronisation	2	2	**4**
Battery*	Replacement with corded power connection	Battery removed	3	3	**9**
Disposal	Disposal instructions	As stated	5	2	**10**
Manufacturing	Dimensional inspection during and after manufacture	As stated	7	1	**7**

Accumulative risk (43) from battery requires further evaluation

PATELLA TRACKER: 3D STEREO-CAMERA MOUNTED TO KNEE BRACE

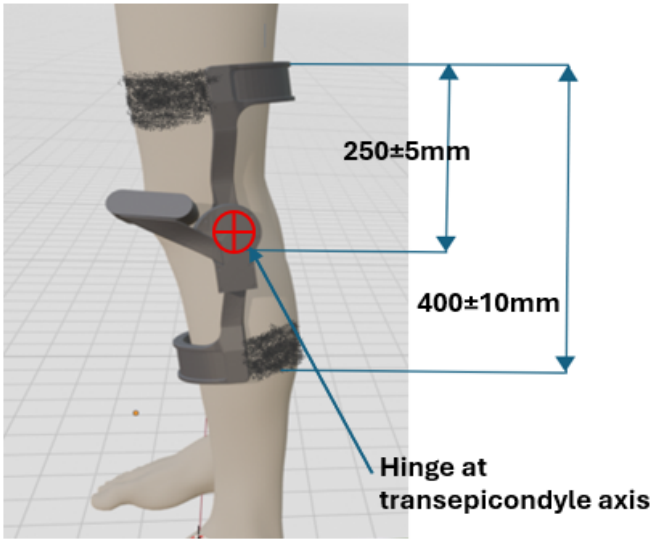






Knee brace allows camera to remain at fixed distance relative to the knee, allowing high accuracy imaging of motion.

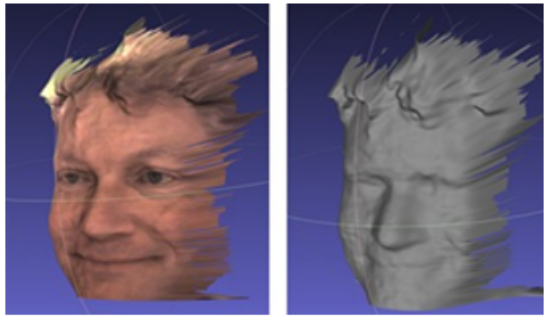
The potentiometer tracks knee flexion angle which is synchronised with 3D camera data.



Intel RealSense® Camera



Stereo camera, produces 3D image data as shown in the image below



3D point cloud at 60 fps, Sub-millimetre accuracy at close range (7cm-50cm)

(Image rights: intel®)

Key design changes following review:

- Battery Removed
- Silicone gasket to provide splash protection
- Filleted edges added to housing

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Appendix:

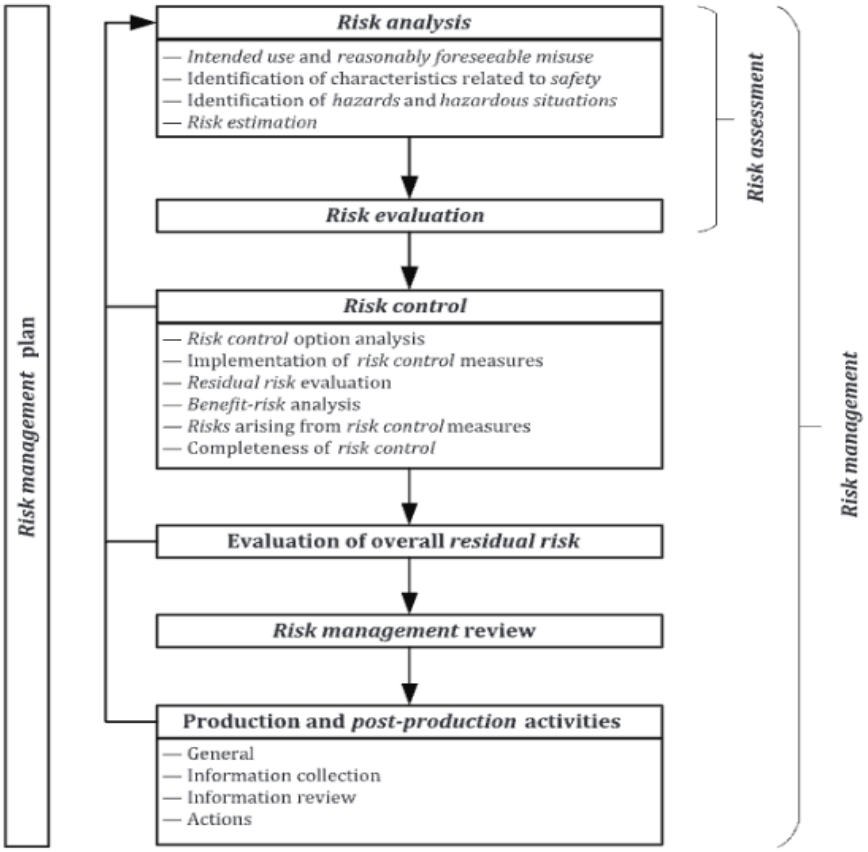


Figure A1. A schematic representation of the risk management process

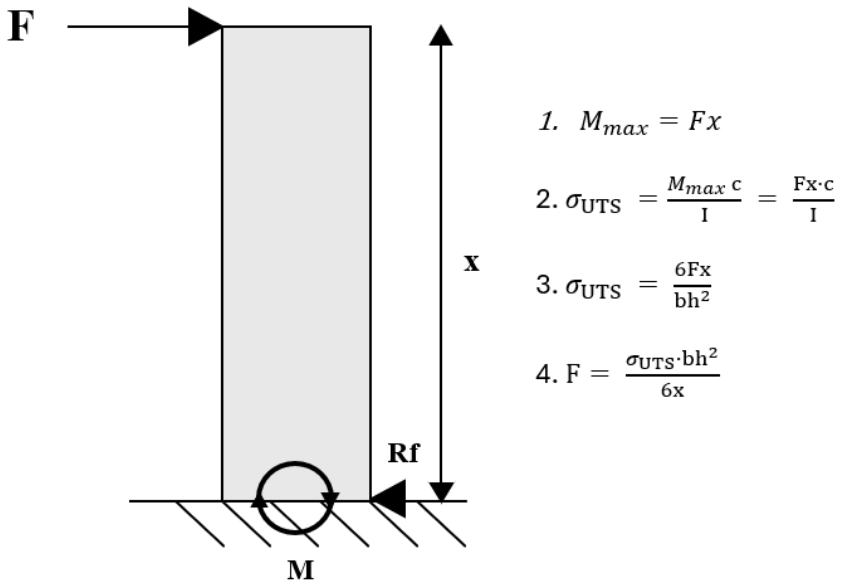


Figure A2. Free body diagram and calculation for main brace support